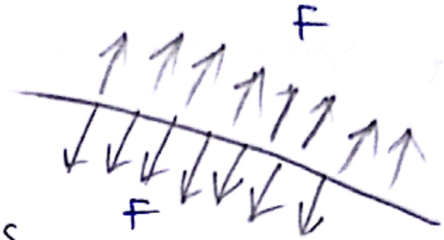
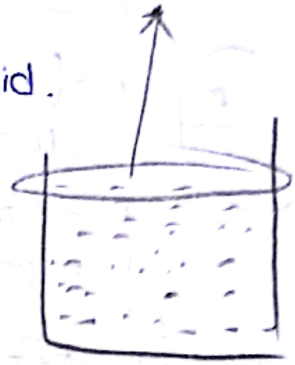
10. Surface Tension ML^0T^{-2}



1. forces.

Surface Tension is a Property Of Fluid.



T = Surface Tension.

F = surface Tension force.

$$\boxed{T = \frac{F}{l}}$$

Hot water : less ST

Cold water : More ST.

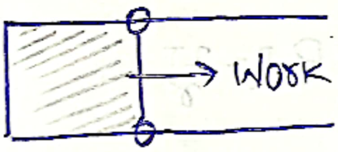
$S\,Tension \propto \frac{1}{T}$

$F = Tl$

where

$l \longrightarrow$ Length Of contact.

2. Surface Energy:



(i) Every Surface will have Surface Energy:

$U = (\text{Surface Area})(T)$.

① liquid drop: Of radius r

$$\boxed{U = 4\pi r^2 T}$$

② Soap Bubble:

$$\boxed{U = 2(4\pi r^2)T}$$

NOTE:

Surface Area increases
Energy is Absorbed.

Surface Area Decreases
Energy is released..

Work Done = Surface Tension $\times$ ΔArea.

18. 1000 water Drops Each radius r forms Bigger Drop

Find Energy Of this Drop

$$1000\left(\cancel{\frac{4}{3}\pi} r^3\right) = \cancel{\frac{4}{3}\pi} R^3$$

$$\boxed{R = 10r}$$

$U = (4\pi r^2 T) \times 1000$ ($\because$ since 1000 Drops)

$U' = (4\pi R^2 T)$

$U' = (4\pi (10r)^2 T)$

$U' = 100U$

$U_i = 4000\pi r^2 T$

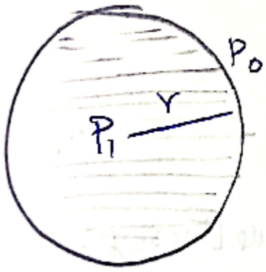
$U_f = 400\pi r^2 T$ $\boxed{\text{Energy released} = 3600\pi r^2 T}$

Units: Surface Tension.

N/m

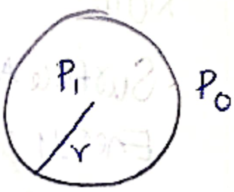
"Excess Pressure:"

1. Excess Pressure inside a liquid drop:



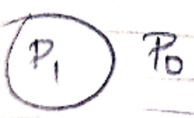
$$\boxed{P_1 - P_0 = \frac{2T}{r}}$$

2. Excess Pressure Inside a Soap bubble:

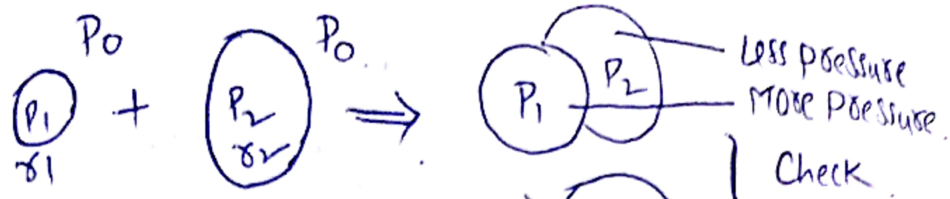


$$\boxed{P_1 - P_0 = \frac{4T}{r}}$$

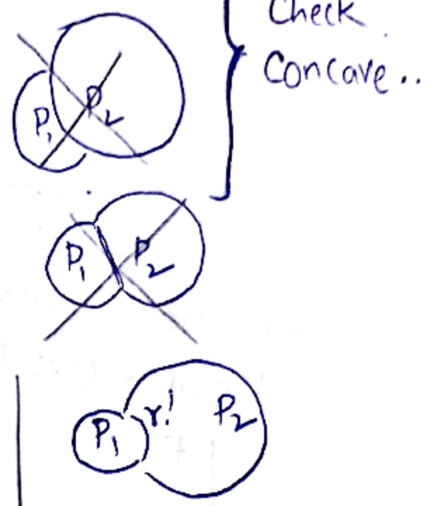
3. Excess Pressure inside an Air bubble formed inside Water.



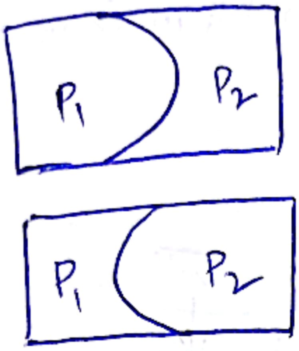
$$P_1 - P_0 = \frac{2T}{r}$$



$P_1 > P_2$ (since radius is less)



Ideal



$P_1 > P_2$

$P_2 > P_1$

$$P_1 - P_0 = \frac{4T}{r_1}$$
$$P_2 - P_0 = \frac{4T}{r_2}$$
$$P_1 - P_2 = 4T\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

NOTE:

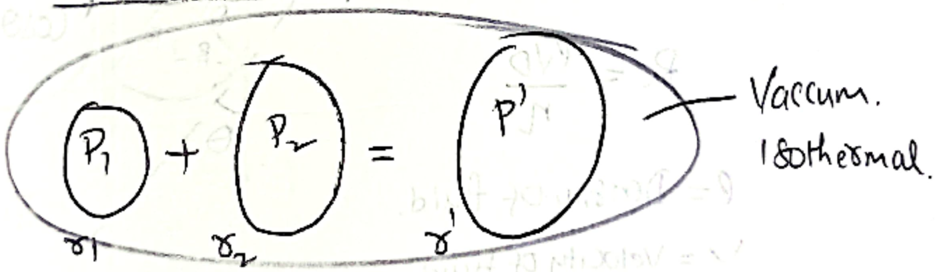
Specific Gravity = Density of Sub / Density of Water

no units...

$$\frac{\cancel{4T}}{r'} = P_1 - P_2 = \cancel{4T}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\boxed{\frac{1}{r'} = \frac{1}{r_1} - \frac{1}{r_2}} \quad (1)$$

2Q.



Get relation b/w r' r_1, r_2

$$n_1 + n_2 = (n_1 + n_2)$$

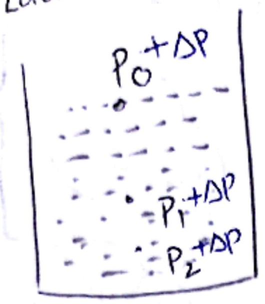
Ideal Gas Equation $\boxed{PV = nRT}$ $\boxed{n = \frac{PV}{RT}}$

$$\frac{P_1V_1}{\cancel{RT}} + \frac{P_2V_2}{\cancel{RT}} = \frac{P'V'}{\cancel{RT}}$$

$$\left(\frac{\cancel{4T}}{r_1}\right)\left(\cancel{\frac{4}{3}\pi} r_1^3\right) + \left(\frac{\cancel{4T}}{r_2}\right)\left(\cancel{\frac{4}{3}\pi} r_2^3\right) = \left(\frac{\cancel{4T}}{r'}\right)\left(\cancel{\frac{4}{3}\pi} r'^3\right)$$

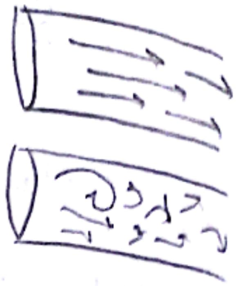
$$r_1^2 + r_2^2 = (r')^2 \quad (2)$$

Pascals Law:

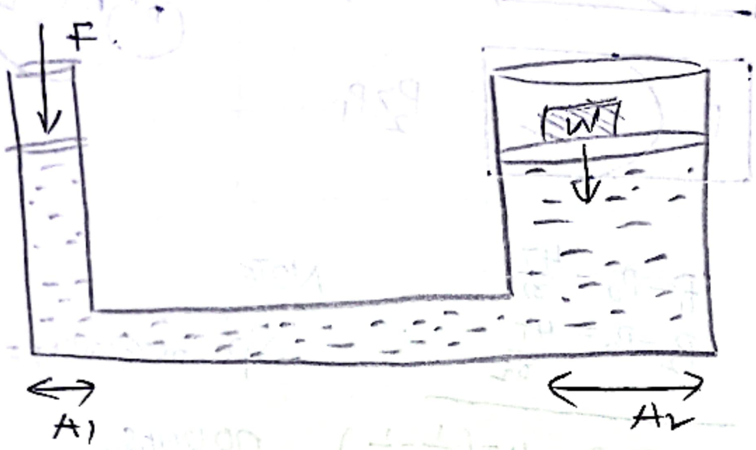


NOTE:

- Laminar flow
- Turbulent flow:



- Change, Variation in Pressure result in Same change, Variation over other level.



$$\frac{F}{A_1} = \frac{W}{A_2} \quad (\because P_1 = P_2 \text{ Application Of Pascal Law})$$

$$W = \left(\frac{A_2}{A_1}\right) F \quad \text{Since } A_2 >> A_1$$

$$W >> F$$

Reynolds number:

$$Re = \frac{\rho V D}{\eta}$$

ρ = Density of fluid

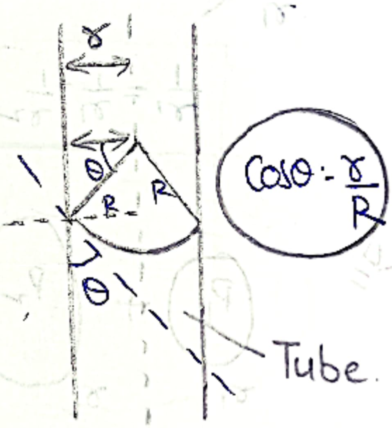
V = Velocity of fluid

D = Diameter of Pipe

η = Coeff of Viscosity

- $Re < 2000$ $\boxed{\text{Laminar}}$

  $Re > 5000$ $\boxed{\text{Turbulent flow}}$



$\cos\theta = \frac{r}{R}$

$T \rightarrow$ Surface Tension

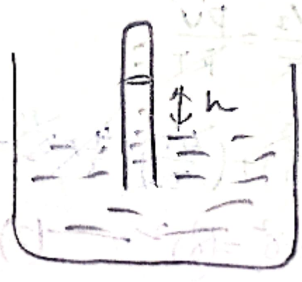
$\rho \rightarrow$ Density of fluid

$\theta \rightarrow$ Contact Angle

$r \rightarrow$ radius of Tube

$R \rightarrow$ radius of Meniscus

Capillary rise:



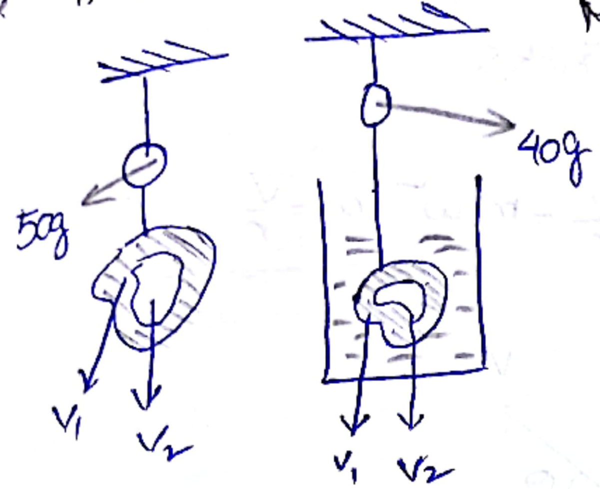
$$h = \frac{2T\cos\theta}{\rho g r}$$

$$h = \frac{2T}{\rho g R}$$

18. An Object with cavity weight 50gr in Air and 40gr in Water If Density of Object is 10g/cc find the Volume of the cavity!

Mass $M_1 = 50$grams

$M_2 = 40$grams.



$V_1 \longrightarrow$ Volume Of Material

$V_2 \longrightarrow$ Volume of cavity...

$V_1 \rho_1 = 50g$

$V_1 = 5 \frac{\cancel{g}}{\cancel{g} cc}$

$\boxed{B = 10}$ ($\because$ Since $W_1 - W_2 = 50 - 40 = 10$).

$(V_1 + V_2)(1) = 10.$

$V_2 = 10 - 5 = 5$